

Method	GSM8K			Humaneval			MMLU-Pro		
	maj@2	maj@4	maj@8	pass@2	pass@4	pass@8	maj@2	maj@4	maj@8
Self-consistency ($\tau = 1$)	19.26	25.32	31.61	18.29	26.22	33.54	13.61	15.15	16.24
Self-consistency ($\tau = 0.6$)	29.57	36.24	41.32	31.10	42.07	48.17	16.15	16.43	17.84
Beam search	35.10	36.47	36.92	31.10	29.88	32.93	15.49	15.29	15.01
Diverse beam search	—	36.47	41.39	—	32.32	33.54	-	16.20	16.58
DeepConf	21.68	26.91	33.36	—	—	—	14.41	15.58	16.76
Self-certainty	20.09	25.32	32.22	—	—	—	14.51	15.92	14.68
Stochastic beam search	33.24	34.82	35.30	31.10	32.98	34.27	14.31	15.38	15.97
Arithmetic sampling	19.82	25.73	32.12	32.03	35.45	35.97	14.32	15.21	16.39
Path-consistency	21.79	26.03	32.74	—	—	—	13.55	14.36	15.87
Self-consistency (top- p &top- k)	25.47	31.84	38.74	31.71	36.59	45.73	14.96	16.66	17.35
DLE (top- p &top- k)	34.80	41.17	44.43	39.02	43.90	51.22	16.36	17.30	18.19
Self-consistency (min- p)	26.46	33.51	39.04	32.93	38.41	46.34	14.99	16.36	17.26
DLE (min- p)	33.74	39.58	43.97	38.41	45.12	53.05	16.45	17.40	17.89
Self-consistency (ε -sampling)	26.84	33.51	40.94	32.93	39.63	47.56	14.89	16.07	17.20
DLE (ε -sampling)	34.57	40.64	44.05	38.41	45.12	52.44	16.64	17.37	17.97

Table 1: **DLE with various samplers improves performance compared to stochastic self-consistency.** Performance (maj@k and pass@k) of various methods on GSM8K, Humaneval and MMLU-Pro with Qwen2.5-0.5B-Instruct. Values are left blank when the method is not applicable to the metric. For all added baselines, the per-step candidate set is the same ε -truncated distribution ($\varepsilon = 0.05$) used by DLE, so the comparison varies the method while holding the sampling distribution fixed.

Stochastic beam search (Gumbel-top- k) [1]. We draw k distinct sequences without replacement using the Gumbel-top- k trick applied to the same ε -truncated next-token distribution as DLE ($\varepsilon = 0.05$). At each step, Gumbel noise is added to the truncated, renormalized log-probabilities, and the k highest perturbed beams are expanded, yielding k exact samples without replacement. We use $k \in \{2, 4, 8\}$ and aggregate with uniform-weighted majority voting (maj@k) or unit-test success (pass@k), matching the DLE and self-consistency protocols. Model, dataset, and prompts are held fixed.

Arithmetic sampling [2]. We sample k sequences in parallel from an arithmetic codebook implicitly defined by the model over the same ε -truncated distribution as DLE ($\varepsilon = 0.05$). Code points are placed on an evenly spaced lattice over the unit interval, $\{(i-0.5)/k : i = 1, \dots, k\}$, which yields the diversity property while keeping decoding embarrassingly parallel. We use $k \in \{2, 4, 8\}$, temperature 1.0, and the same aggregation (uniform-weighted majority voting / pass@k) as the other methods. Unlike Gumbel-top- k , arithmetic sampling does not guarantee strictly distinct sequences in finite samples; it spreads samples across the distribution while remaining an unbiased sampler from the model.

Path-consistency [3]. We generate k paths per question and, after each path, extract the highest-confidence answer prefix to guide the generation of subsequent branches, following the

prefix-enhancement procedure of the original work. To isolate the prefix-reuse mechanism from the choice of sampler, we use the same ε -truncated distribution ($\varepsilon = 0.05$) as DLE and self-consistency. We sweep the confidence threshold over $\{0.6, 0.7, 0.8, 0.9\}$ and report the best setting at $k \in \{2, 4, 8\}$; a threshold of 1.0 recovers standard self-consistency. Because path-consistency relies on confidence over an extracted answer to select prefixes, it targets answer-voting (maj@k) tasks and is not evaluated on the pass@k metric.

References

- [1] Wouter Kool, Herke van Hoof, and Max Welling. Stochastic Beams and Where to Find Them: The Gumbel-Top-k Trick for Sampling Sequences Without Replacement. *arXiv preprint arXiv:1903.06059*, 2019.
- [2] Luke Vilnis, Yury Zemlyanskiy, Patrick Murray, Alexandre Tachard Passos, and Sumit Sanghai. Arithmetic sampling: parallel diverse decoding for large language models. In *International Conference on Machine Learning*, 2023.
- [3] Jiace Zhu, Yingtao Shen, Jie Zhao, and An Zou. Path-consistency: Prefix enhancement for efficient inference in llm. *arXiv preprint arXiv:2409.01281*, 2024.